WHEATSTONE BRIDGE CIRCUIT

1. Definition

A Wheatstone bridge is an electrical circuit used to measure an unknown resistance accurately by balancing two arms of a bridge circuit. It consists of four resistances, a galvanometer, and a battery.

2. Construction

A Wheatstone bridge consists of the following components:

1. **Four resistors, P, Q, R and S, connected in the form of a bridge.**
2. **A galvanometer (G) connected between the two middle junctions.**
3. **A battery connected across the other two opposite junctions.**
4. **A variable resistor used to balance the bridge.**

3. Principle of Wheatstone Bridge

The Wheatstone bridge works on the principle of null deflection. When the bridge is balanced, no current flows through the galvanometer because the potential difference between its two terminals is zero.

4. Balance Condition

Let P, Q, R and S be the four resistances of the bridge.

When the bridge is balanced:

$P/Q = R/S$

Therefore,

$PS = QR$

If S is the unknown resistance, then:

$S = QR/P$

This equation is used to calculate the unknown resistance.

5. Working

1. **The battery supplies current to the bridge circuit.**
2. **Current flows through the four resistance arms.**
3. **The variable resistance is adjusted until the galvanometer shows zero deflection.**
4. **At this point, the bridge is balanced and the potential at the two galvanometer terminals is equal.**
5. **The unknown resistance is calculated using the balance condition.**

6. Derivation of Balance Condition

Let the four resistances be P, Q, R and S.

At the balanced condition, the galvanometer current is zero.

Therefore, the potential difference across P equals the potential difference across R:

$I_1P = I_2R$ ...(1)

Similarly, the potential difference across Q equals the potential difference across S:

$I_1Q = I_2S$ ...(2)

Dividing equation (1) by equation (2):

P/Q = R/S

Therefore,

PS = QR

Hence, the balance condition of the Wheatstone bridge is:

P/Q = R/S

7. Applications

1. **Used to measure unknown electrical resistance accurately.**
2. **Used in strain gauge circuits to measure mechanical strain.**
3. **Used in temperature measurement circuits with resistance temperature sensors.**
4. **Used in light-sensitive and other resistive sensor circuits.**
5. **Used in electrical measuring instruments.**

8. Advantages

1. **Provides accurate measurement of unknown resistance.**
2. **Uses a null-deflection method, reducing errors associated with direct current measurement.**
3. **The balance condition is independent of the battery voltage for an ideal circuit.**
4. **Suitable for measuring a wide range of moderate resistance values.**

9. Limitations

1. **Lead and contact resistances can introduce errors when measuring very low resistances.**
2. **Very high resistances can be difficult to measure accurately because of detector sensitivity and leakage currents.**
3. **Heating due to excessive current may change the resistance values.**
4. **Accurate balancing requires suitable resistors and a sensitive galvanometer.**

10. Conclusion

The Wheatstone bridge is an important electrical circuit used to determine unknown resistance. It operates on the principle of null deflection, and at balance, the ratio of resistances in one pair of arms equals the ratio in the other pair.